## NAME

**mbotps** − Predicts tides using tidal prediction methods and data derived from the OSU Tidal Prediction Software (OTPS) distribution.

## VERSION

Version 5.0

## SYNOPSIS

**mbotps**
[
**--input**=*datalist* {**-I***datalist*}
**--format**=*format_id* {**-F***format_id*}
**--use-mbprocess** {**-M**}
**--skip-existing** {**-S**}
**--tide-output**=*file* {**-O***output*}
**--tide-position**=*lon/lat* {**-R***lon/lat*}
**--tide-format**=*format_id* {**-A***tideformat*}
**--interval**=*interval* {**-D***interval*}
**--start-time**=*year/month/day/hour/minute/second* {**-B***year/month/day/hour/minute/second*}
**--end-time**=*year/month/day/hour/minute/second* {**-E***year/month/day/hour/minute/second*}
**--tide-station-file**=*file* {**-N***tidestationfile*}
**--tide-station-position**=*lon/lat {***-U***lon/lat}*
**--tide-station-format**=*format_id {***-C***tidestationformat}*
**--otps-model**=*model_name {***-T***model}*
**--otps-path**=*path {***-P***path}*
**--help** *{***-H***}*
**--verbose** *{***-V***}*
*]*

## DESCRIPTION

**MBotps** is a utility that predicts tides using tidal prediction methods and global tidal models developed by Gary Egbert and Lana Erofeeva of Oregon State University (OSU), whose OSU Tidal Prediction Software (OTPS) and TPXO tidal models are available at:
https://www.tpxo.net

Unlike earlier versions of this program (still available, deprecated, as **mbotpsold**(1)), **mbotps** does not install, build, or run any part of the OTPS Fortran software. Instead it implements the same harmonic tidal prediction methods natively - equilibrium tidal arguments and nodal corrections, inference of minor constituents from major ones, and bilinear interpolation of the tidal atlas grids - reading OTPS "atlas" format tide model binary files directly. This removes the need to separately obtain, install, and compile OTPS's predict_tide program, which previously required a working Fortran compiler and was one of the more error-prone steps in setting up **mbotps**.

Users must still separately obtain and install one or more OTPS tidal models (the binary model data files themselves); these remain outside the **MB-System** installation, at the location given by the **−−otps-path** option. Model data are made available by OSU at no cost for academic, research, or other noncommercial use, but must be licensed for a fee for commercial use. Instructions for obtaining and installing a tidal model are given below.

**MBotps** can be operated in two modes. First, users may use the **−R**, **−B**, and **−E** options to specify a location and the beginning and end times of a tidal model for that location. The **−D** option sets the time interval of

values in the output tidal model, and the **−O** option sets the output tidal model filename.

Alternatively, instead of specifying a place and time range, the user may specify one or more swath data files using the **−I** option. A tidal model is produced for each swath file in which tidal values are calculated using the sonar navigation locations at intervals specified with the **−D** option, and if the **−M** option is specified, the swath file's processing parameter file is modified so that **mbprocess** applies the tidal model during processing.

The **−C***tidestationformat*, **−N***tidestationfile*, and **−U***tidestationlon/tidestationlat* commands together allow users to input observations from a tide station; these observations can be used to calculate corrections to tidal model values in the vicinity of the tide station. If tide station data are specified, then **MBotps** calculates the difference between the observed and modeled tide at that station for each data point in the input tide station data. This difference time series is then used as a correction to the output tide models, whether at a location specified with the **−R***lon/lat* option or for swath data specified with the **−I***datalist* option.

## ATTRIBUTION AND LICENSING OF TIDAL PREDICTION METHODS

The tidal prediction methods used by **mbotps** - equilibrium tidal arguments and nodal corrections, inference of minor tidal constituents from major ones, and the OTPS binary tidal atlas file format itself - are derived from the OSU Tidal Prediction Software (OTPS), authored by:

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https://www.tpxo.net/

The equilibrium argument and nodal correction formulas, and the minor-constituent inference method, were originally supplied to OTPS by Richard Ray (NASA/GSFC).

OTPS (both its source code and the TPXO tidal model data it works with) is copyrighted by Oregon State University and distributed under a royalty-free, nonexclusive license for academic, research, and other noncommercial use, which permits modification and redistribution of the software subject to conditions including attribution of the original copyright; see OTPS's own COPYRIGHT file for the complete license text. **MB-System**'s native implementation of these methods (in mb_otps_predict.c and mb_otps_predict.h) is a transcription and reimplementation of OTPS's numerical methods in C, not a copy of the OTPS Fortran source code, and is provided as a derivative work under that same OSU license.

## OBTAINING AND INSTALLING A TPXO TIDAL MODEL

**mbotps** reads OTPS "atlas" format tidal model binary data files directly; it does not require the OTPS software distribution or a Fortran compiler to be installed at all. Only a tidal model's data files are needed, obtained separately from OSU.

A tidal model can be requested from OSU via the web pages:

https://www.tpxo.net/global
https://www.tpxo.net/tpxo-products-and-registration

Basically, one must request access to a model via email. If approved, you will receive a link to download a directory named e.g. TPXO10_atlas_v2 which contains files named:

grid_tpxo10_atlas_30_v2 u_2n2_tpxo10_atlas_30_v2
h_2n2_tpxo10_atlas_30_v2 u_k1_tpxo10_atlas_30_v2
h_k1_tpxo10_atlas_30_v2 u_k2_tpxo10_atlas_30_v2
h_k2_tpxo10_atlas_30_v2 u_m2_tpxo10_atlas_30_v2
h_m2_tpxo10_atlas_30_v2 u_m4_tpxo10_atlas_30_v2
h_m4_tpxo10_atlas_30_v2 u_mf_tpxo10_atlas_30_v2
h_mf_tpxo10_atlas_30_v2 u_mm_tpxo10_atlas_30_v2
h_mm_tpxo10_atlas_30_v2 u_mn4_tpxo10_atlas_30_v2
h_mn4_tpxo10_atlas_30_v2 u_ms4_tpxo10_atlas_30_v2

h_ms4_tpxo10_atlas_30_v2 u_n2_tpxo10_atlas_30_v2
h_n2_tpxo10_atlas_30_v2 u_o1_tpxo10_atlas_30_v2
h_o1_tpxo10_atlas_30_v2 u_p1_tpxo10_atlas_30_v2
h_p1_tpxo10_atlas_30_v2 u_q1_tpxo10_atlas_30_v2
h_q1_tpxo10_atlas_30_v2 u_s1_tpxo10_atlas_30_v2
h_s1_tpxo10_atlas_30_v2 u_s2_tpxo10_atlas_30_v2
h_s2_tpxo10_atlas_30_v2

(**mbotps** only ever reads the "h_*" elevation files; the "u_*" current/transport files and the "grid_*" bathymetry file are not used and need not be present.)

**mbotps** expects tidal models to be organized under a directory structure matching the OTPS convention: a top-level directory (the **−−otps-path** location, default /usr/local/src/otps/) containing a "DATA" subdirectory. Move the downloaded model directory into place there, e.g.:

mv TPXO10_atlas_v2 /usr/local/src/otps/DATA/

That alone is enough for **mbotps** to find and use the model: pass the model directory's own name as **−−otps-model** (e.g. **−−otps-model=TPXO10_atlas_v2**), and **mbotps** will scan it directly for the "h_*" elevation files it needs. No "Model_"-prefixed control file is required.

Alternatively, a "Model_"-prefixed control file (e.g. "Model_tpxo10_atlas") may be used instead, exactly as with the original OTPS software: this is a small text file, one per installed model, containing three lines naming the elevation, transport, and bathymetry files using a "*" in place of the constituent name, e.g.:

DATA/TPXO10_atlas_v2/h_*_tpxo10_atlas_30_v2
DATA/TPXO10_atlas_v2/u_*_tpxo10_atlas_30_v2
DATA/TPXO10_atlas_v2/grid_tpxo10_atlas_30_v2

Such a control file, if present, takes precedence over scanning the model directory directly, and lets **−−otps-model** be set to a short name (e.g. "tpxo10_atlas") that need not match the model directory's own name. This is mainly useful for compatibility with an existing OTPS installation that already has these files (they are part of the OTPS software distribution, not the tidal model data downloads).

This can be tested by running **mbotps** with the −−help option:

Program mbotps
MB-system Version 5.8.3beta16

MBotps predicts tides using methods and data derived from the OSU Tidal Prediction Software (OTPS) distributions.

usage: mbotps
[
--input=datalist {-Idatalist}
--format=format_id {-Fformat_id}
--use-mbprocess {-M}
--skip-existing {-S}
--tide-output=file {-Ooutput}
--tide-position=lon/lat {-Rlon/lat}
--tide-format=format_id {-Atideformat}
--interval=interval {-Dinterval}
--start-time=year/month/day/hour/minute/second {-Byear/month/day/hour/minute/second}
--end-time=year/month/day/hour/minute/second {-Eyear/month/day/hour/minute/second}
--tide-station-file=file {-Ntidestationfile}
--tide-station-position=lon/lat {-Ulon/lat}
--tide-station-format=format_id {-Ctidestationformat}
--otps-model=model_name {-Tmodel}

```
--otps-path=path {-Ppath}
--help {-H}
--verbose {-V}
]
```

```
Checking for available OTPS tide models
  OTPS location: /usr/local/src/otps
  Default OTPS model name: tpxo10_atlas
  Specified OTPS model name:
  Possible OTPS tidal models:
    tpxo10_atlas <installed>
  Number of available OTPS tide models: 1

Using OTPS tide model:  tpxo10_atlas
Opened OTPS atlas model 'tpxo10_atlas': 10800 x 5401 grid, 15 constituents: m2 s2 k1 o1 n2 p1 k2 q1 2n2
mf mm m4 ms4 mn4 s1
```

The output of **mbotps** indicates that the desired tidal model is available and was opened directly (no external predict_tide program is invoked).

## MB-SYSTEM AUTHORSHIP

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Monterey Bay Aquarium Research Institute
Dale N. Chayes
Center for Coastal and Ocean Mapping
University of New Hampshire
Christian do Santos Ferreira
MARUM - Center for Marine Environmental Sciences
University of Bremen

## OPTIONS

−−**input**=*datalist*

Sets the input filename. If *format* > 0 (set with the −−**format** option) then the swath sonar data contained in *infile* is read and processed. If *format* < 0, then *infile* is assumed to be an ascii file containing a list of the input swath sonar data files to be processed and their formats. The program will read the data in each one of these files. In the *infile* file, each data file should be followed by a data format identifier, e.g.:

```
datafile1 11
datafile2 24
```

This program uses the **MBIO** library and will read or write any swath sonar format supported by **MBIO**. A list of the swath sonar data formats currently supported by **MBIO** and their identifier values is given in the **MBIO** manual page. Default: *infile* = "stdin".

−−**format**=*format_id*

Sets the data format of the input swath data file specified with the −**I** option. If *format* < 0, then the input file specified with the −**I** option will actually contain a list of input swath sonar data files. This program uses the **MBIO** library and will read any swath sonar format with timestamps supported by **MBIO**. A list of the swath sonar data formats currently supported by **MBIO** and their identifier values is given in the **MBIO** manual page. The default format is set using **mbdefaults**.

−−**use-mbprocess**
If the −**I** option has been used so that tidal models are created for swath files, then this option causes each swath file's parameter file to be modified so that **mbprocess** will read and apply the ancillary tidal model file created by **mbotps**.

−−**skip-existing**
If tide models are being generated for swath files specified using the **-I** option, then skip files that already have an existing tide model.

−−**tide-output=***file*
Sets the filename of the tidal model output.

−−**tide-position=***lon/lat*
Sets the longitude and latitude position at which the tidal model will be calculated.

−−**tide-format=***format_id*
This option sets the format used to write the output tidal model. If *format_id*=2, each output line has the format <yr mon day hour min sec tide>. Any other value (*format_id*=1, 3, or 4) produces output lines with the format <time_d tide>, where time_d is the decimal seconds since 1/1/1970. Default: *format_id* = 2.

−−**interval**=*interval*
This option sets the time interval between tidal model values in seconds. Default: 60 seconds.

−−**start-time**=*year/month/day/hour/minute/second*
This option sets the starting time for for the output tidal model.

−−**end-time**=*year/month/day/hour/minute/second*
This option sets the ending time for the output tidal model

−−**tide-station-file**=*file*
Sets the filename of the tide station data file used to correct the output tide model.

−−**tide-station-position=***lon/lat*
Sets the longitude and latitude position of the tide station from which the data specified with the −−**tide-station-file** option were obtained.

−−**tide-station-format**=*format_id*
This option sets the format of the tide station data in the file specified using the **-N***tidestationfile* option. The tide station data may be in one of four ASCII, space delimited, table formats:
*tidestationformat*=1: format is <time_d tide>
*tidestationformat*=2: format is <yr mon day hour min sec tide>
*tidestationformat*=3: format is <yr jday hour min sec tide>
*tidestationformat*=4: format is <yr jday daymin sec tide>
Note that in format 1 the value time_d = decimal seconds since 1/1/1970 and that format 4 the value daymin = decimal minutes since the start of day.

−−**otps-model**=*model_name*
Sets the name of the OTPS tidal model to be used. This model must be part of the local OTPS model installation at the location given by −−**otps-path**, either as a "Model_"-prefixed control file (e.g. "tpxo9_atlas_v5" for "Model_tpxo9_atlas_v5") or as the name of a model directory found directly under DATA/ (e.g. "TPXO10_atlas_v2"); see OBTAINING AND INSTALLING A TPXO TIDAL MODEL above. Default: "tpxo10_atlas"

−−**otps-path**=*path*
Sets the path to the local installation of OTPS tidal model data, i.e. the directory containing the "DATA" subdirectory with the "Model_"-prefixed control files and the tidal model data files themselves. Default: /usr/local/src/otps

**−−help**
This "help" flag cause the program to print out a description of its operation and then exit immediately.

**−−verbose**
Increases the verbosity of **mbotps**.

## EXAMPLES

Suppose one wishes to obtain a tidal model for the location 121W 36N extending at 60 second intervals over the day February 21, 2009. The following will suffice:

```
mbotps −−tide-position=125/36 −−start-time=2009/02/21/00/00/00 \
                    −−end-time=2009/02/21/23/59/59 −−verbose
```

The shell output looks like:

```
Program mbotps
MB-system Version 5.8.3beta16

Checking for available OTPS tide models
  OTPS location: /usr/local/src/otps
  Default OTPS model name: tpxo10_atlas
  Specified OTPS model name:
  Possible OTPS tidal models:
    tpxo10_atlas <installed>
  Number of available OTPS tide models: 1

Using OTPS tide model:  tpxo10_atlas
Opened OTPS atlas model 'tpxo10_atlas': 10800 x 5401 grid, 15 constituents: m2 s2 k1 o1 n2 p1 k2 q1 2n2 mf mm m4 ms4 mn4 s1

Results are in tide_model.txt
```

The output tidal model is in the file tide_model.txt, which includes data that look like:

```
# Tide model generated by program mbotps
# MB-System Version: 5.8.3beta16
# Tide model generated by program mbotps
# using tidal prediction methods and data derived from the
# OSU Tidal Prediction Software (OTPS) distribution, see:
#     https://www.tpxo.net/
#
# OTPS tide model:
#      tpxo10_atlas
# Constituents included: m2  s2  k1  o1  n2  p1  k2  q1  2n2 mf  mm  m4  ms4 mn4 s1
# Output format:
#      year month day hour minute second tide
# where tide is in meters
# Run by user <caress> on cpu <RAITT-1121.local> at <Tue Nov 19 21:36:30 2024>
2009 02 21 00 00 00   -0.6780
2009 02 21 00 01 00   -0.6760
2009 02 21 00 02 00   -0.6730
2009 02 21 00 03 00   -0.6710
2009 02 21 00 04 00   -0.6680
2009 02 21 00 05 00   -0.6650
```

```
2009 02 21 00 06 00   -0.6630
2009 02 21 00 07 00   -0.6600
2009 02 21 00 08 00   -0.6580
2009 02 21 00 09 00   -0.6550
2009 02 21 00 10 00   -0.6520
2009 02 21 00 11 00   -0.6490
2009 02 21 00 12 00   -0.6470
2009 02 21 00 13 00   -0.6440
 ..................
2009 02 21 23 54 00   -0.7900
2009 02 21 23 55 00   -0.7890
2009 02 21 23 56 00   -0.7870
2009 02 21 23 57 00   -0.7860
2009 02 21 23 58 00   -0.7840
2009 02 21 23 59 00   -0.7820
```

Now, suppose that one wants to apply tide corrections directly to a set of EM3002 data in GSF format. First, execute **mbotps** with the datalist for the swath data specified as input:

```
mbotps −−input=datalist.mb-1 −−verbose
```

The resulting shell output looks like:

```
Program mbotps
MB-system Version 5.8.3beta16

Checking for available OTPS tide models
 OTPS location: /usr/local/src/otps
 Default OTPS model name: tpxo10_atlas
 Specified OTPS model name:
 Possible OTPS tidal models:
        tpxo10_atlas <installed>
 Number of available OTPS tide models: 1

Using OTPS tide model:  tpxo10_atlas
Opened OTPS atlas model 'tpxo10_atlas': 10800 x 5401 grid, 15 constituents: m2 s2 k1 o1 n2 p1 k2
q1 2n2 mf mm m4 ms4 mn4 s1

Model tide for swath data referenced by datalist.mb-1

himbb05291.d23.mb121 : model tide at 412 of 35602 records
```

The output tide files have the same structure shown above:

```
# Tide model generated by program mbotps
# MB-System Version: 5.8.3beta16
# Tide model generated by program mbotps
# using tidal prediction methods and data derived from the
# OSU Tidal Prediction Software (OTPS) distribution, see:
#    https://www.tpxo.net/
#
# OTPS tide model:
#     tpxo10_atlas
# Constituents included: m2  s2  k1  o1  n2  p1  k2  q1  2n2 mf  mm  m4  ms4 mn4 s1
# Output format:
#     year month day hour minute second tide
```

```
# where tide is in meters
# Run by user <caress> on cpu <deitz> at <Tue Nov 19 17:53:22 2024>
2005 10 18 19 01 36   0.0800
2005 10 18 19 02 36   0.0790
2005 10 18 19 03 36   0.0770
2005 10 18 19 04 36   0.0760
2005 10 18 19 05 37   0.0750
2005 10 18 19 06 37   0.0730
2005 10 18 19 07 37   0.0720
2005 10 18 19 08 37   0.0710
2005 10 18 19 09 37   0.0700
```

In addition to generating *.tde files for each swath file referenced by the input datalist structure, when the **−−use−mbprocess** option is specified, **mbotps** modifies the parameter file associated with each swath file (creating it if necessary) so that tide correction is enabled using the *.tde file and tide format 2. When **mbprocess** is run on the same datalist, the files will be reprocessed, and the processing will include the application of the tide correction to all bathymetry.

## SEE ALSO

**mbsystem**(1), **mbotpsold**(1), **mbprocess**, **mbset**

## BUGS

None known. If **mbotps** cannot find a usable tidal model, obtaining and installing one is somewhat involved (see OBTAINING AND INSTALLING A TPXO TIDAL MODEL above), since it requires registering with OSU to download TPXO tidal model data. The deprecated **mbotpsold** is retained (in src/deprecated, built only with --enable-deprecated or -DbuildDeprecated=ON) as a fallback and as a cross-check against the original OTPS Fortran reference implementation (predict_tide), for users who have that program already installed.